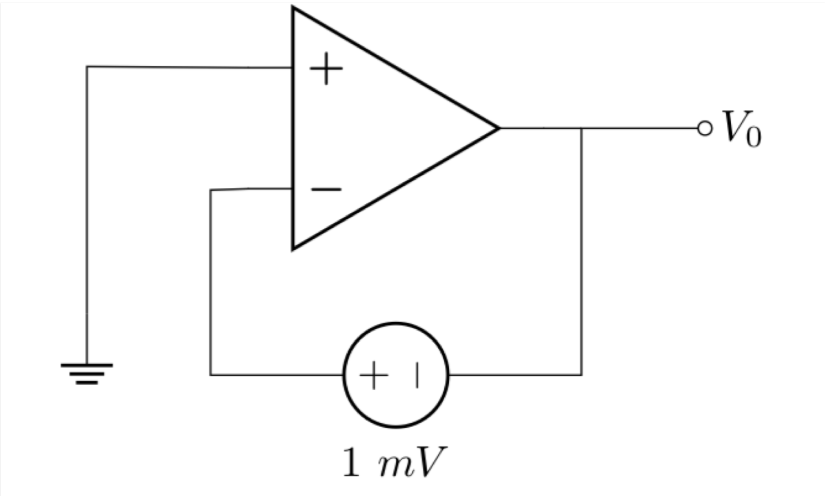
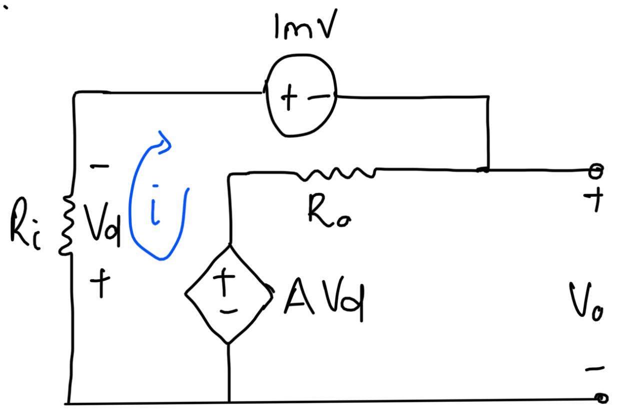
[redacted]

Tutorial 6

5.6 Determine V_o if this is a 741 Opamp (it is not ideal)



Redraw as:



$A = 2\times10^5$

$R_i = 2\ M\Omega$

$R_o = 50\ \Omega$

$V_d = R_i \cdot i$

$= 2\times10^6\ i$

KVL:

$$2\times10^6\ i + 1\times10^{-3} + 50\ i + 2\times10^5\ V_d = 0$$

$$i\left[2\times10^6 + 50 + (2\times10^5)(2\times10^6)\right] = -1\times10^{-3}$$

$$i = -24.99987\times10^{-16}$$

$$= -2.5\ fA$$

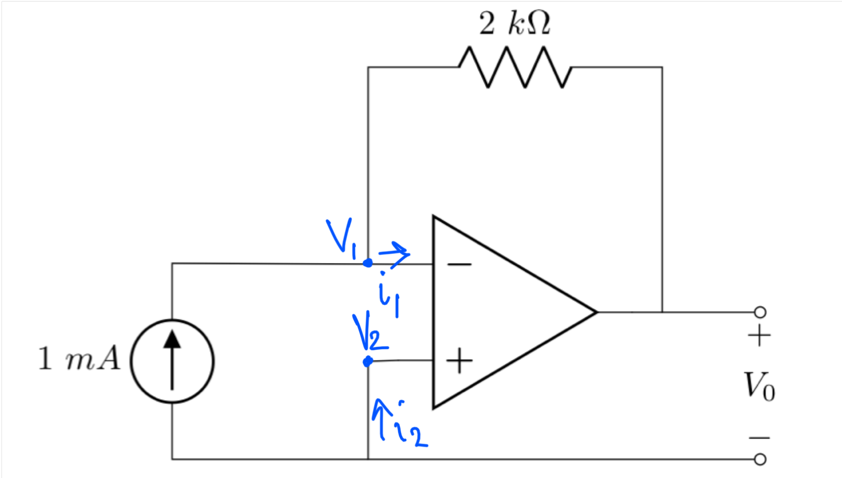
$$-AV_d + iR_o + V_o = 0$$

$$V_o = (2\times10^5)(2\times10^6)(-2.5\times10^{-15}) - (-2.5\times10^{-15})(50)$$

$$= -1\ mV$$

5.8 Determine V_o

(a)



$\Rightarrow$ Ideal op-amp

$i_1 = 0$; $i_2 = 0$

$V_1 = V_2$

$V_1 = V_2 = 0V$

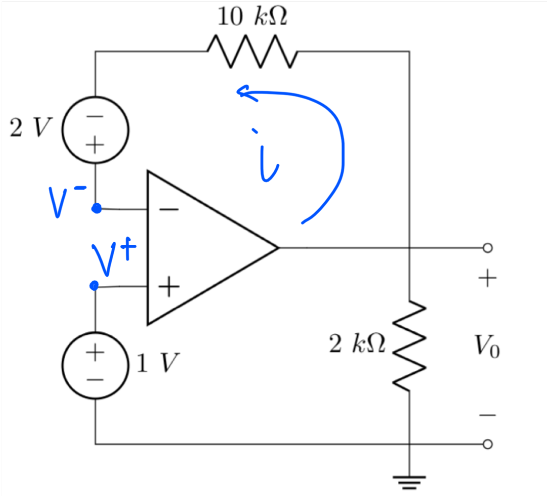
@ Node V_1

$$-1 \times 10^{-3} + \frac{V_1 - V_o}{2000} = 0$$

$$\frac{-V_o}{2000} = 1 \times 10^{-3}$$

$$V_o = -2 \text{ V}$$

(b)



$$V_1 = V_2 = 1\,V$$

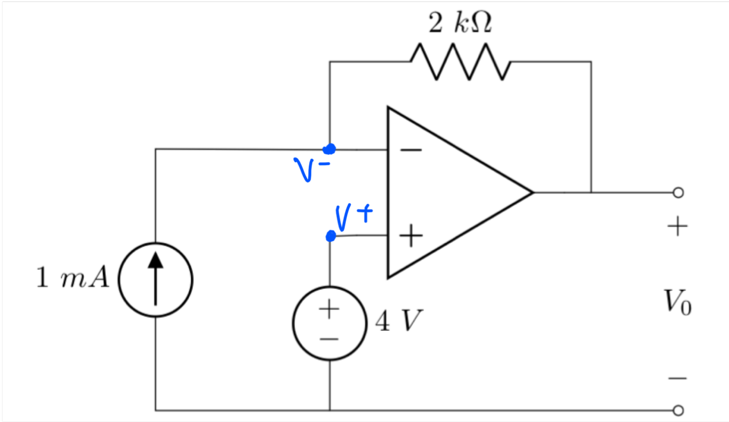
For an ideal op-amp $i = 0\,A$

$$\therefore V_{10k\Omega} = 0\,(10\,000)$$
$$= 0\,V$$

$$V^- = V_o + 2$$
$$V_o = 1 - 2$$
$$= -1\,V$$

5.9 Determine V_o

(a)



Ideal op-amp:

$$V^+ = V^- = 4V$$

Node V^-:

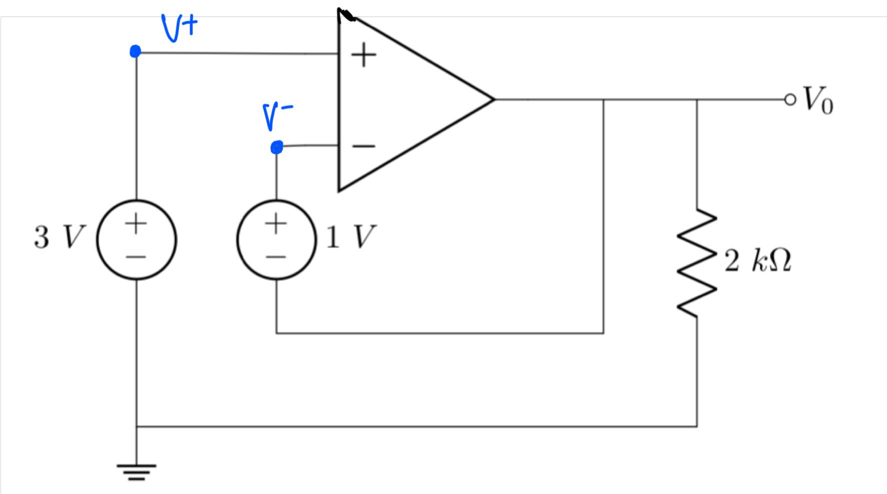
$$-1\times10^{-3} + \frac{V^- - V_o}{2000} = 0$$

$$4 - V_o = 2$$

$$-V_o = 2-4$$

$$V_o = 2\ V$$

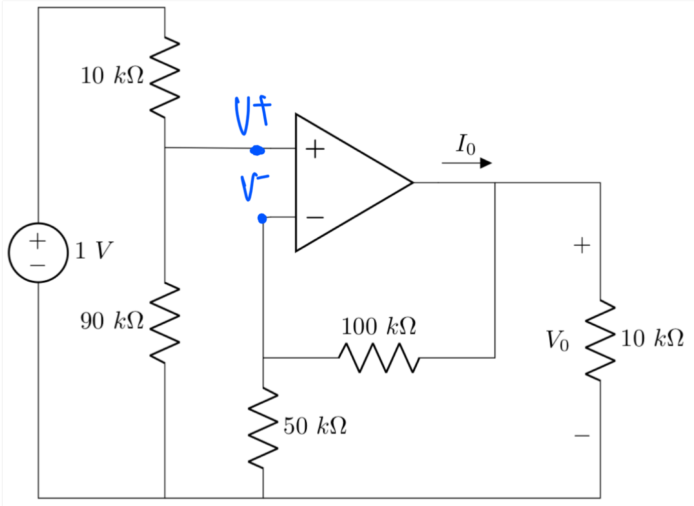
(b)



$V^{+} = V^{-} = 3V$

$$V_o = V^{-} - 1$$
$$= 3 - 1$$
$$= 2V$$

5.13 Determine I_o



Ideal op-amp:

⇒ Voltage division

$$V^+ = 1 \times \frac{90\,000}{90\,000 + 10\,000}$$

$$= 0.9 \text{ V}$$

$$V^+ = V^-$$

⇒ Nodal @ V^-

$$\frac{V^- - V_o}{100\,000} + \frac{V^-}{50\,000} = 0$$

$$\frac{0.9 - V_o}{100\,000} + \frac{0.9}{50\,000} = 0$$

$$V_o = 2.7 \text{ V}$$

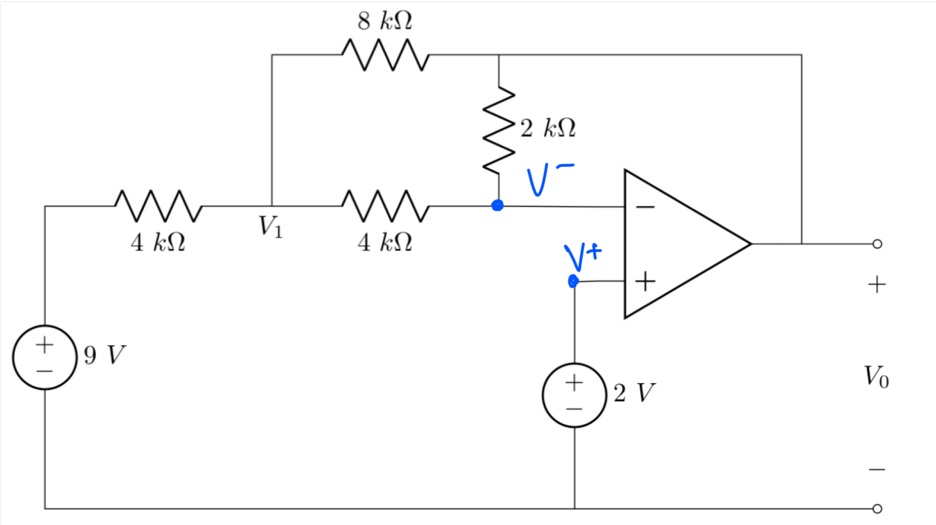
$\Rightarrow$ Nodal @ V_o

$$-I_o + \frac{V_o - V^-}{100\,000} + \frac{V_o}{10\,000} = 0$$

$$I_o = \frac{2.7 - 0.9}{100\,000} + \frac{2.7}{10000}$$

$$= 288\ \mu A$$

5.20 Determine V_o



⇒ Ideal op-amp

$$V^+ = V^- = 2\,V$$

⇒ Nodal analysis node V_1

$$\frac{V_1 - 9}{4000} + \frac{V_1 - V^-}{4000} + \frac{V_1 - V_o}{8000} = 0$$

$$5V_1 - V_o = 22$$

⇒ Nodal analysis node V^-

$$\frac{V^- - V_1}{4000} + \frac{V^- - V_o}{2000} = 0$$

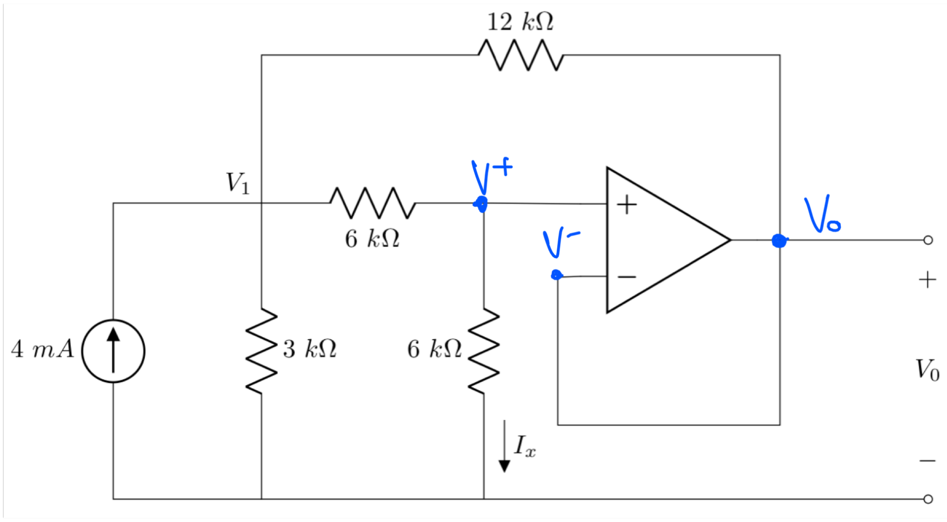
$$V_1 + 2V_o = 6$$

⇒ Solve

$$V_o = 0.727\ V$$

5.31 (a) Determine V_o

(b) Determine I_x



$V^- = V_o$

$V^+ = V^- = V_o$

$\Rightarrow$ Modal at V_1 :

$$\frac{V_1}{3000} + \frac{V_1 - V^+}{6000} + \frac{V_1 - V_o}{12\,000} - 4\times10^{-3} = 0$$

$$\frac{V_1}{3000} + \frac{V_1 - V_o}{6000} + \frac{V_1 - V_o}{12\,000} = 4\times10^{-3}$$

$$7V_1 - 3V_o = 48 \qquad (1)$$

$\Rightarrow$ Modal at V^+ :

$$\frac{V^+ - V_1}{6000} + \frac{V^+}{6000} = 0$$

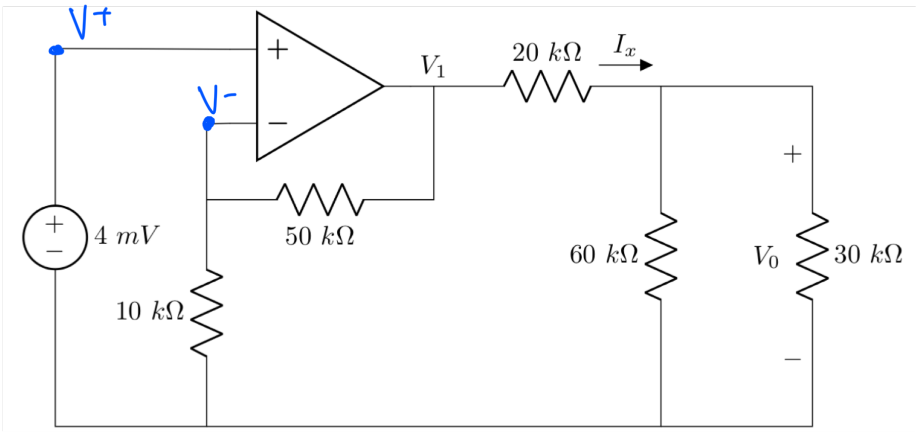
$$V_0 - V_1 + V_0 = 0$$
$$2V_0 = V_1 \qquad (2)$$

$\Rightarrow$ (2) in (1)

$$7(2V_0) - 3V_0 = 48$$
$$14V_0 - 3V_0 = 48$$
$$V_0 = 4.3636\ V$$

$$\Rightarrow I_x = \frac{V^+}{6000}$$
$$= \frac{4.3636}{6000}$$
$$= 727.2727\ \mu A$$

5.32 Determine I_x and V_o



Ideal op-amp

$V^+ = V^- = 4mV$

⇒ Modal at V^-

$$\frac{4\times10^{-3} - V_1}{50\,000} + \frac{4\times10^{-3}}{10\,000} = 0$$

$V_1 = 24mV$

⇒ Modal at V_o

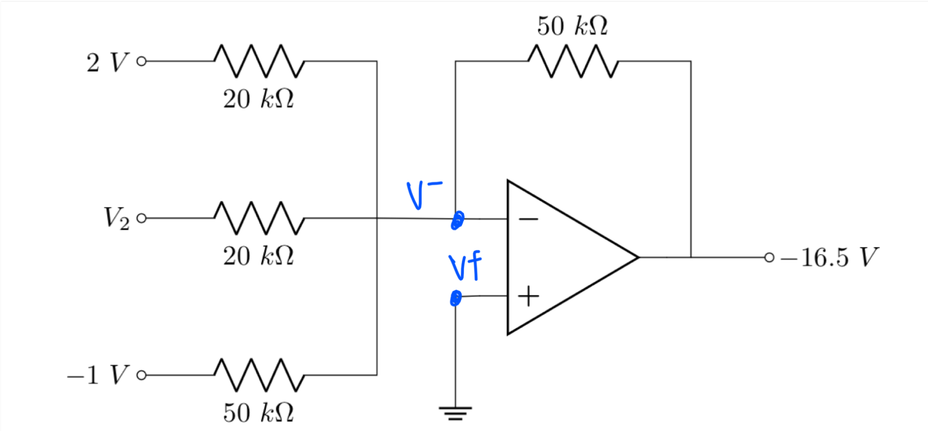
$$\frac{V_o}{30000} + \frac{V_o}{60\,000} + \frac{V_o - V_1}{20\,000} = 0$$

$V_o = 12\ mV$

⇒ $I_x = \dfrac{V_1 - V_o}{26\,000}$

$$= \frac{24\times10^{-3} - 12\times10^{-3}}{20\times10^{3}} = 0.6\ \mu A$$

5.39 Determine V_{TH} and R_{TH} across a-b



Ideal op-amp:

$V^+ = V^- = 0V$

⇒ Nodal at V^-

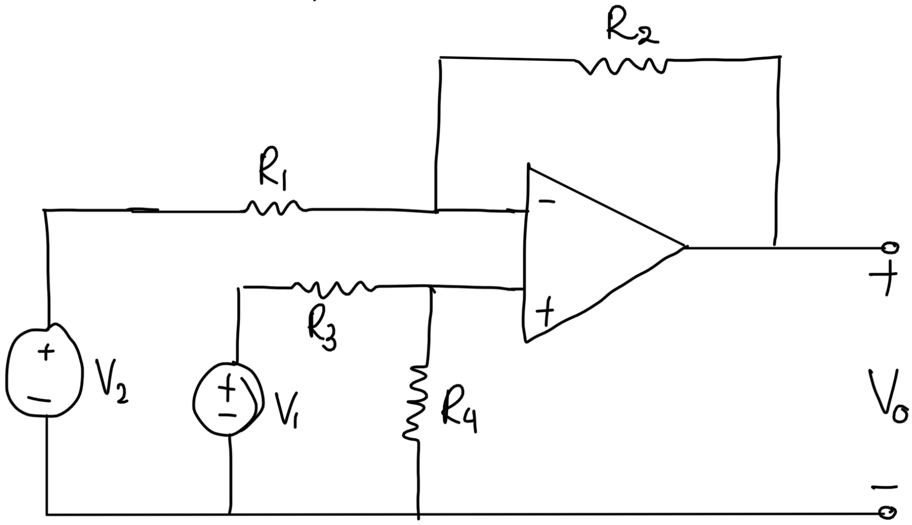
$$\frac{V^- - (-16.5)}{50\,000} + \frac{V^- - (-1)}{50\,000} + \frac{V^- - 2}{20\,000} + \frac{V^- - V_2}{20\,000} = 0$$

$$\frac{16.5}{50\,000} + \frac{1}{50\,000} - \frac{2}{20\,000} - \frac{V_2}{20\,000} = 0$$

$V_2 = 5\,V$

5.45 Design an op-amp circuit with the following I-O relationship: $V_o = 3V_1 - 2V_2$

$\Rightarrow$ Only known op-amp circuit that does this is the differential amplifier.



Use Eq. 5.18 with V_1 and V_2 swapped:

$$V_o = \frac{R_2\left(1 + \frac{R_1}{R_2}\right)}{R_1\left(1 + \frac{R_3}{R_4}\right)} V_1 - \frac{R_2}{R_1} V_2 \qquad (2)$$

$$V_o = 3V_1 - 2V_2 \qquad (1)$$

$$\frac{R_2}{R_1} = 2 \qquad (3) \qquad\qquad \frac{R_2\left(1 + \frac{R_1}{R_2}\right)}{R_1\left(1 + \frac{R_3}{R_4}\right)} = 3 \qquad (4)$$

Select any values for R_1 and R_4

$\rightsquigarrow$ Let $R_1 = 1k\Omega$ $\quad R_4 = 1k\Omega$

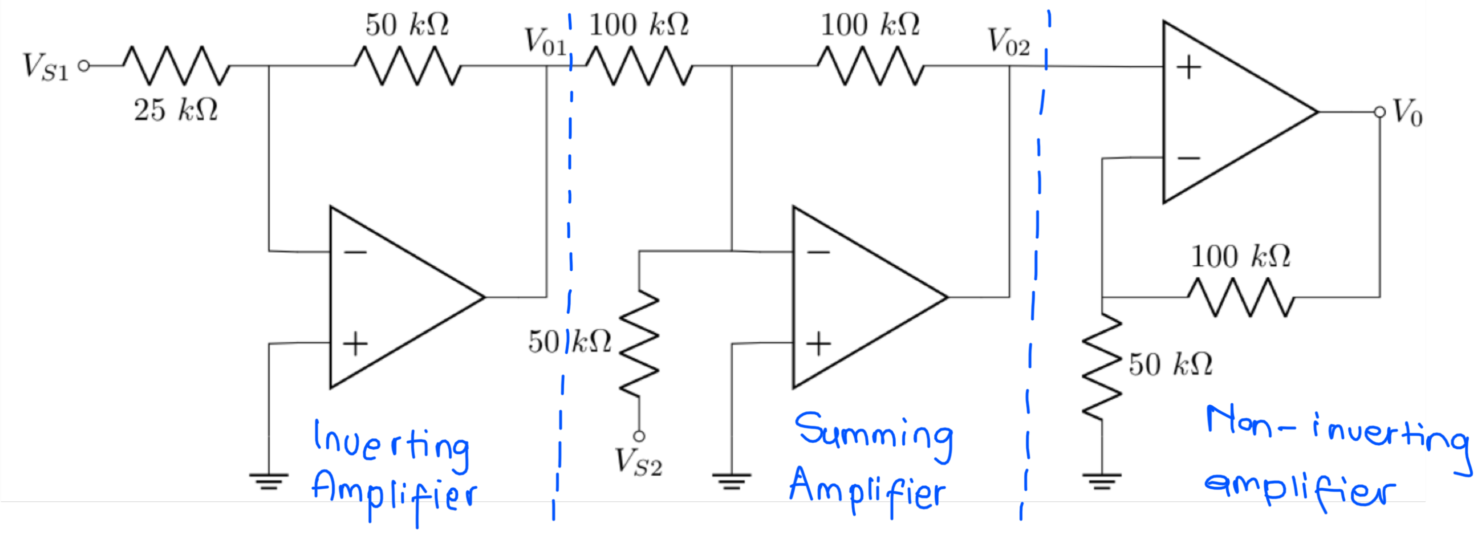
3 $\therefore$ $R_2 = 2R_1$

$= 2k\Omega$

4
$$\frac{2\times10^3\left(1+\frac{1\times10^3}{2\times10^3}\right)}{1\times10^3\left(1+\frac{R_3}{1\times10^3}\right)} = 3$$

$R_3 = 0$

5.57 Determine V_o



⇒ Inverting Amplifier

$$V_{o1} = -\frac{50\,000}{25\,000} \cdot V_{s1}$$

$$= -2\,V_{s1}$$

⇒ Summing Amplifier

$$V_{o2} = -\left(\frac{100}{100}(-2V_{s1}) + \frac{100}{50}(V_{s2})\right)$$

$$= -(-2V_{s1} + 2V_{s2})$$

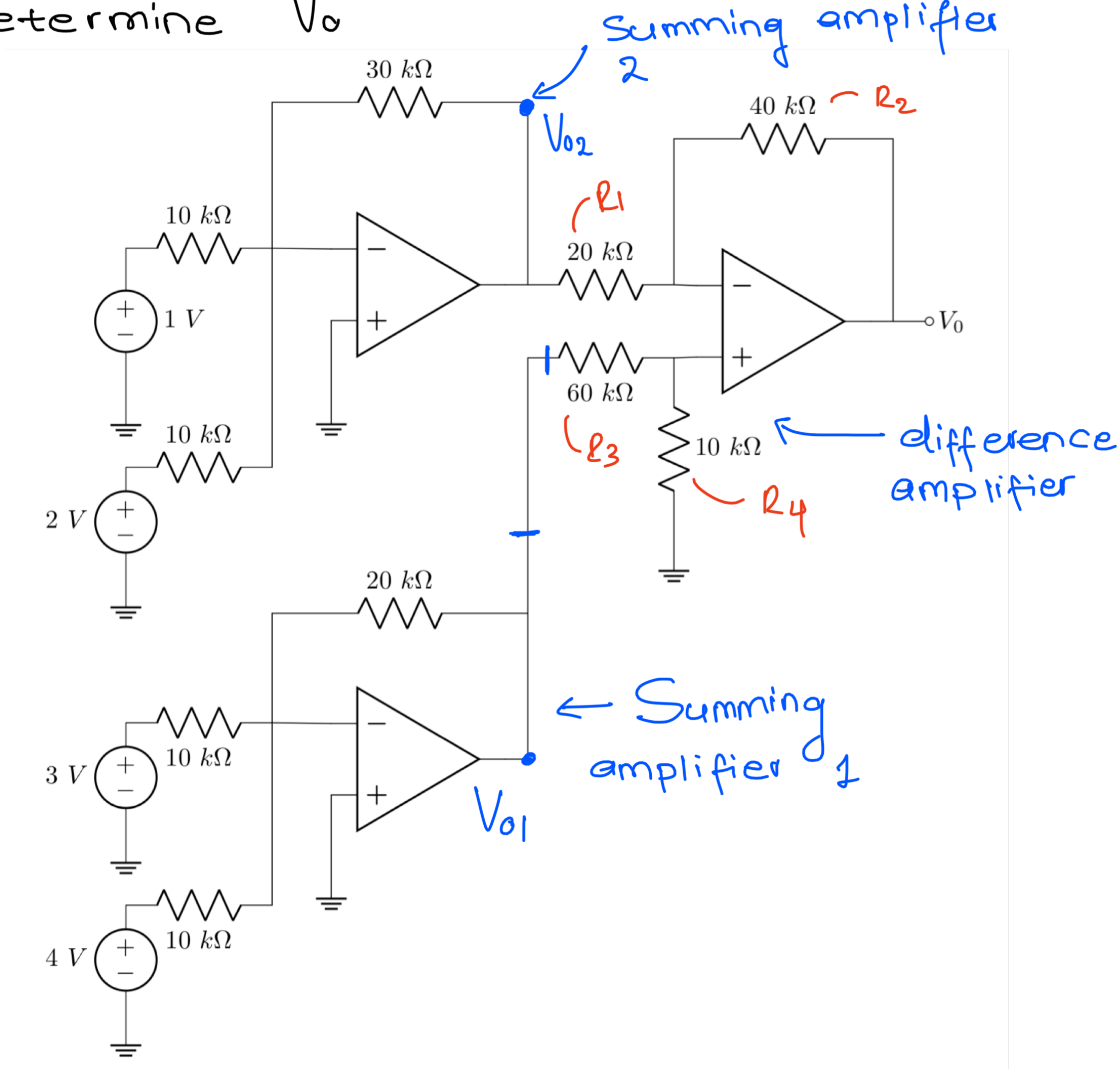
$$= 2V_{s1} - 2V_{s2}$$

⇒ Non-inverting amplifier

$$V_o = \left(1 + \frac{100}{50}\right)(2V_{s1} - 2V_{s2})$$

$$= 6\,(V_{s1} - V_{s2})$$

5.70 Determine V_o



$\Rightarrow$ Summing amplifier 1

$$V_{o1} = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)$$

$$= -\left(\frac{20\times10^3}{10\times10^3}(3) + \frac{20\times10^3}{10\times10^3}(4)\right)$$

$$= -14 \text{ V}$$

$\Rightarrow$ Summing amplifier 2

$$V_{o2} = -\left(\frac{30\times10^3}{10\times10^3}(1) + \frac{30\times10^3}{10\times10^3}(2)\right)$$

$$= -9\ V$$

$\Rightarrow$ Difference amplifier

$$V_o = \frac{R_2\left(1+\frac{R_1}{R_2}\right)}{R_1\left(1+\frac{R_3}{R_4}\right)} \cdot V_{o1} - \frac{R_2}{R_1} \cdot V_{o2}$$

$$= \frac{40\times10^3\left(1+\frac{20\times10^3}{40\times10^3}\right)}{20\times10^3\left(1+\frac{60\times10^3}{10\times10^3}\right)} \cdot (-14) - \frac{40\times10^3}{20\times10^3} \cdot (-9)$$

$$= 12\ V$$